

MATRIX-FREE STOCHASTIC CALCULATION OF OPERATOR NORMS WITHOUT USING ADJOINTS

ON THE WAY TO COMPUTE THE ADJOINT MISMATCH



Universität
Bremen

26th Conference of the ILAS 2025

Kaohsiung, Taiwan

Jonas Bresch

Coauthors: D. A. Lorenz, F. Schneppe, M. Winkler

TABLE OF CONTENTS

I. MOTIVATION & CONTRIBUTIONS

II. REMARKS

III. DETAILS

IV. NUMERICAL RESULTS

V. OUTLOOK

- Operator norms are needed in many places, e.g. optimization:
 - Lipschitz constant of the objective $\|Ax - y\|^2$,
 - gradient descent or the forward-backward method need a stepsize $0 < \tau < 2/\|A\|^2$ for convergence^[1],
 - Chambolle-Pock method^[2] minimizing $f(x) + g(Ax)$ for convex, closed and proper functions f and g ,
 - needs stepsize σ, τ that fulfill $\sigma\tau \leq 1/\|A\|^2$ ^[3].
- Solvers: power method, requires A^* , if A is not self-adjoint,
- in discretized setting for $\|A\|$: the adjoint of the discretized one is not the actual adjoint.

[1] Ernest K Ryu and Wotao Yin. *Large-scale convex optimization: algorithms & analyses via monotone operators*. Cambridge University Press, 2022.

[2] Antonin Chambolle and Thomas Pock. “A first-order primal-dual algorithm for convex problems with applications to imaging”. In: *Journal of mathematical imaging and vision* 40 (2011), pp. 120–145.

[3] Kristian Bredies et al. “Degenerate preconditioned proximal point algorithms”. In: *SIAM Journal on Optimization* 32.3 (2022), pp. 2376–2401.

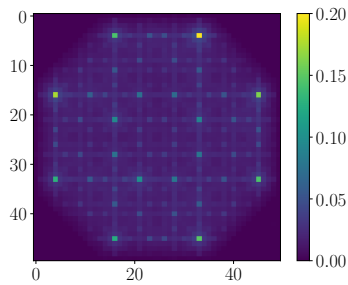
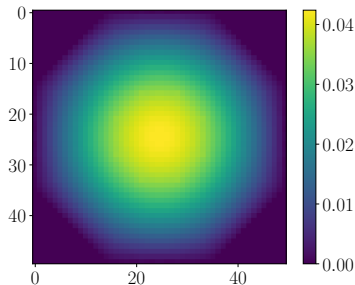
MOTIVATION II—TOY EXAMPLE

- Rotations of a function $f : \mathbb{D} := \{x \in \mathbb{R}^2 : \|x\| \leq 1\} \rightarrow \mathbb{R}$ in $\mathbb{R}^2$,
- given by $A_\alpha f(x) = f(R_\alpha x)$, where

$$R_\alpha := \begin{pmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{pmatrix}.$$

- It holds: $A_\alpha^* = A_{-\alpha}$ and $A_\alpha^{-1} = A_{-\alpha}$, such that $\|A_\alpha\| = 1$ in the continuous setting.

n	α	$\sqrt{\ \mathbf{A}_{-\alpha}\mathbf{A}_\alpha\ }$	$\ \mathbf{A}_\alpha\ $
25	10	0.99999	1.00463
25	30	1.00002	1.05930
25	45	0.99996	1.17387
50	10	1.00080	0.99804
50	30	0.99998	1.05314
50	45	1.00000	1.16840



- Operator norm of a linear map between finite dimensional Hilbert spaces, i.e. $\{m, d\}$.
- Under certain assumption:
 - Evaluations of the linear map $v \mapsto Av$ are available (black-box).
 - Restrictive storage assumptions, i.e. $\mathcal{O}(\max(m, d))$.
 - No access to the adjoint operator A^* .
 - Approximate the value $\|A\|$ to any accuracy.
 - Compute a corresponding maximal singular vector on the fly.

- We propose:
 - Stochastic algorithm that produces a sequence that converges to the operator norm
$$\max_{\|v\|=1} \|Av\|^2.$$
 - Uses evaluations of the operator A , and calculation of norms and inner products.
 - For any linear map between finite Hilbert spaces,
 - do not need any assumptions on the dimension of the spaces or on the structure of A ^[4].
 - does not based on the Schatten- p norm^[5], for which we need $p \rightarrow \infty$, or for computation p sufficiently large enough.
 - Produces an approximation of a right singular vector for the largest singular value,
 - extended to compute further right singular vectors.
 - Convergence result for the corresponding eigenvalue and eigenvector equation,

[4] Weihao Kong and Gregory Valiant. “Spectrum estimation from samples”. In: *Ann. Statist.* 45.5 (2017), pp. 2218–2247. DOI: doi.org/10.1214/16-AOS1525.

[5] Yi Li, Huy L Nguyen, and David P Woodruff. “On sketching matrix norms and the top singular vector”. In: *Proceedings of the twenty-fifth annual ACM-SIAM symposium on Discrete algorithms*. SIAM. 2014, pp. 1562–1581.

- Random search method,
- with exact, i.e. optimal, step size:
 - 1) Initialize some $v^0 \in \mathbb{R}^d$ with $\|v^0\| = 1$ and set $k = 0$.
 - 2) Generate $y^k \sim \mathcal{N}(0, I_d)$.
 - 3) Project y^k onto the space orthogonal to v_k and normalize it

$$x^k = \frac{y^k - \langle y^k, v^k \rangle v^k}{\|y^k - \langle y^k, v^k \rangle v^k\|}$$

to get a random normalized search direction perpendicular to v^k .

- 4) Calculate a step size τ_k by

$$\tau_k \in \arg \max_{\tau} \frac{\|Av^k + \tau Ax^k\|^2}{\|v^k + \tau x^k\|^2}.$$

- 5) Update $v^{k+1} = \frac{v^k + \tau_k x^k}{\|v^k + \tau_k x^k\|}$ and go to step 2.

→ **operates fairly simple** if (4) has an exact, and simple to compute, solution.

TABLE OF CONTENTS

I. MOTIVATION & CONTRIBUTIONS

II. REMARKS

III. DETAILS

IV. NUMERICAL RESULTS

V. OUTLOOK

- By design we always have that $\|v^k\| = \|x^k\| = 1$ and $\langle v^k, x^k \rangle = 0$.
 - The random vector x^k is normally distributed on the tangent space on the unit sphere $\mathbb{S}^{d-1}$ at v^k .
 - By Rayleigh's principle: critical points of the problem $\max_{\|v\|=1} \|Av\|$ are exactly the right singular vectors of A ,
 - The intersection of the eigen space of the largest (or smallest) singular value gives exactly the global maxima (or minima).
 - All other eigen spaces lead to saddle points.
- **Crucially:** We show that the method will (a.s.) avoid all saddle points and will converge (a.s.) to the global maximum, i.e. the operator norm.

TABLE OF CONTENTS

I. MOTIVATION & CONTRIBUTIONS

II. REMARKS

III. DETAILS

IV. NUMERICAL RESULTS

V. OUTLOOK

- Wirst observations hold by construction

$$\|v^k + \tau x^k\|^2 = \|v^k\|^2 + 2\tau \langle v^k, x^k \rangle + \tau^2 \|x^k\|^2 = 1 + \tau^2.$$

- The objective in the maximization problem reads as

$$h_k(\tau) := \frac{\|Av^k + \tau Ax^k\|^2}{1 + \tau^2},$$

- and derivative

$$h'_k(\tau) = 2 \frac{a_k + b_k \tau - a_k \tau^2}{(1 + \tau^2)^2},$$

- where

$$a_k := \langle Av^k, Ax^k \rangle \quad \text{and} \quad b_k := \|Ax^k\|^2 - \|Av^k\|^2.$$

- **All parameters** are available under our certain restrictions.

SOLVING STEP SIZE PROBLEM

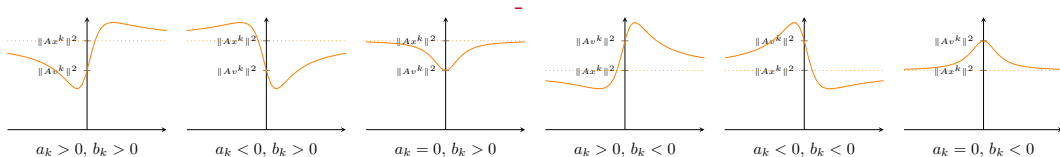
PROPOSITION 1 (CLOSED FORM—STEPSIZE)

If $a_k \neq 0$, then the maximizer of h_k (the optimal step size from v^k in direction x^k) is unique and given by

$$\tau_k = \text{sign}(a_k) \left(\frac{b_k}{2|a_k|} + \sqrt{\frac{b_k^2}{4a_k^2} + 1} \right).$$

If $a_k = 0$, then we have:

$$\{\tau_k = 0 : b_k < 0, \quad \tau_k \rightarrow \pm\infty : b_k > 0, \quad \text{entire } \mathbb{R} : b_k = 0$$



DISCUSS THE CASES FOR a_k

- As long as $a_k \neq 0$, the algorithm leads to a strictly increasing sequence $\|Av^k\|$.
- Convergence follows, since the sequence is bounded by $\|A\|$.
- Our method is a stochastic projected gradient method:
 - $\tau_k > 0$ if and only if $a_k = \langle Av^k, Ax^k \rangle = \langle A^*Av^k, x^k \rangle > 0$,
 - hence, we move in direction $\text{sign}(a_k)x_k$,
 - direction is uniformly distributed on a half-sphere in the space orthogonal to v^k ,
 - by symmetry: $\mathbb{E}(\text{sign}(a_k)x^k)$ is a positive multiple of the projection of A^*Av^k on the space orthogonal to v^k .

LEMMA 1

Assume: v^k is not a right singular vector of $A : \mathbb{R}^d \rightarrow \mathbb{R}^m$.

$$\Rightarrow a_k = \langle Av^k, Ax^k \rangle \neq 0 \text{ a.s..}$$

THEORETICAL ASSUMPTIONS ON A

LEMMA 2

Assume:

- σ^2 is an eigenvalue of $A^*A : \mathbb{R}^d \rightarrow \mathbb{R}^d$ with geometric multiplicity $r < d$ and
- $v \notin E = \{x \in \mathbb{R}^d \mid A^*Ax = \sigma^2x\}$ the respective eigen space.

$\Rightarrow \{x \in \{v\}^\perp \mid v + x \in E\}$ is an affine subspace of dimension r or $r - 1$.

COROLLARY 3

Assume: all singular values of $A : \mathbb{R}^d \rightarrow \mathbb{R}^m$ have multiplicities less than $d - 1$.

If v^k is not a right singular vector $\Rightarrow v^{k+1}$ is a.s. also not a right singular vector.

ASSUMPTION 1

The linear operator $A : \mathbb{R}^d \rightarrow \mathbb{R}^m$ has no singular value with multiplicity $d - 1$ or d .

→ **Consequently:** $a_k \neq 0$ (a.s.) throughout the iteration if v^0 is not a right singular vector of A and the sequence $\|Av^k\|$ is increasing and convergent (a.s.).

- We summarize some basic properties one can prove: For v^k, x^k and a_k holds
 - $\|Av^{k+1}\|^2 - \|Av^k\|^2 = \tau_k a_k \rightarrow 0$ a.s. for $k \rightarrow \infty$,
 - $a_k = \langle Av^k, Ax^k \rangle \rightarrow 0$ a.s. for $k \rightarrow \infty$,
 - $\mathbb{E}[a_k^2 \mid v^k] = \frac{1}{d-1} \|(I_d - v^k(v^k)^*)A^*Av^k\|^2 \rightarrow 0$ a.s. for $k \rightarrow \infty$.

PROPOSITION 2

Every accumulation point of the sequence v^k is a right singular vector of A a.s..

- Consider the set $B := \{v \in \mathbb{S}^{d-1} : \|Av\|^2 > \sigma_2^2\}$, it holds
 - If $v^k \in B$, then $v^{k+1} \in B$,
 - If $v^{k_0} \in B$ holds for some k_0 , then $\lim_{k \rightarrow \infty} \|Av^k\| = \sigma_1 = \|A\|$ a.s.
- **Technically:** we just need $v^{k_0} \in B$ for the final a.s. convergence!

ALMOST SURE CONVERGENCE II

- Let $\varepsilon > 0$ and $v, v^* \in \mathbb{S}^{d-1}$ and define

$$D_v := \left\{ x \in \{v\}^\perp \mid \frac{v + \tau x}{\|v + \tau x\|} \in B_\varepsilon(\{\pm v^*\}) \text{ s.t. } \exists \tau \in \mathbb{R} \right\},$$

- denote by $\mathcal{U}_v$ the uniform distribution on the unit sphere in $\{v\}^\perp$.

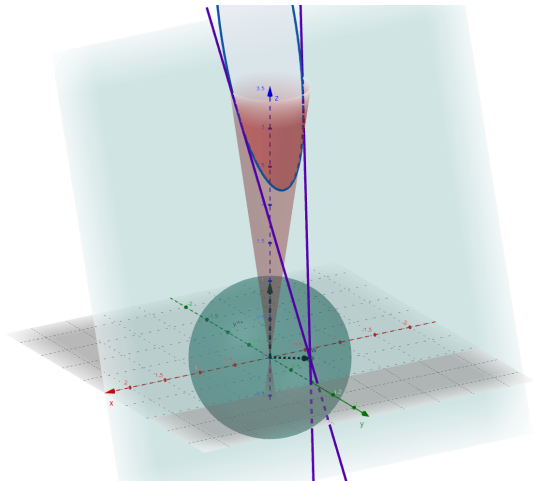
LEMMA 4

There exists a positive constant $p_{\varepsilon,d}$ such that

$$\inf_{v \in \mathbb{S}^{d-1}} \mathcal{U}_v(D_v) \geq p_{\varepsilon,d}.$$

THEOREM 5

- It holds $\lim_{k \rightarrow \infty} \|Av^k\| = \sigma_1$ a.s..
- If the geometric multiplicity of σ_1 is one, then v^k converges a.s., too.



- Stopping criteria: given by $a_k \rightarrow 0$,
- combining with $\mathbb{E}[a_k^2 \mid v^k] = \frac{1}{d-1} \|A^* A v^k - \|A v^k\|^2 v^k\|^2$.
- Two singular values: If one singular value has geometrical multiplicity $d - 1$, then we either have
 - $\|A v^0\| > \sigma_2$ if the smallest is of multiplicity $d - 1$, and $a_k \neq 0$ by construction,
 - again $\|A v^0\| > \sigma_2$ if the largest is of multiplicity $d - 1$, now, we sample in an $d - 1$ dimensional subspace, which is exactly the one corresponding to the largest eigenvector.
- One singular value: we have the following equivalence

$$\mathbb{P}[a_0 = 0] = 1 \quad \Leftrightarrow \quad \mathbb{P}[a_0 = 0] > 0 \quad \Leftrightarrow \quad A^* A = c I_d.$$

- Study the corresponding singular vector and singular value equation

$$\|A^*Av^k - \|Av^k\|^2v^k\|^2, \quad k \in \mathbb{N}, \quad \text{where } v^k \text{ is from our method.}$$

- We show a sublinear rates in different settings.

PROPOSITION 3

It holds a.s. $\min_{0 \leq k \leq n} a_k^2 \leq \frac{2\|A\|^4}{n+1}.$

THEOREM 6

It holds a.s. $\min_{0 \leq k \leq n} \|A^*Av^k - \|Av^k\|^2v^k\|^2 \leq \frac{2(d+1)\|A\|^4}{n+1}.$

TABLE OF CONTENTS

I. MOTIVATION & CONTRIBUTIONS

II. REMARKS

III. DETAILS

IV. NUMERICAL RESULTS

V. OUTLOOK

PROOF OF CONCEPT

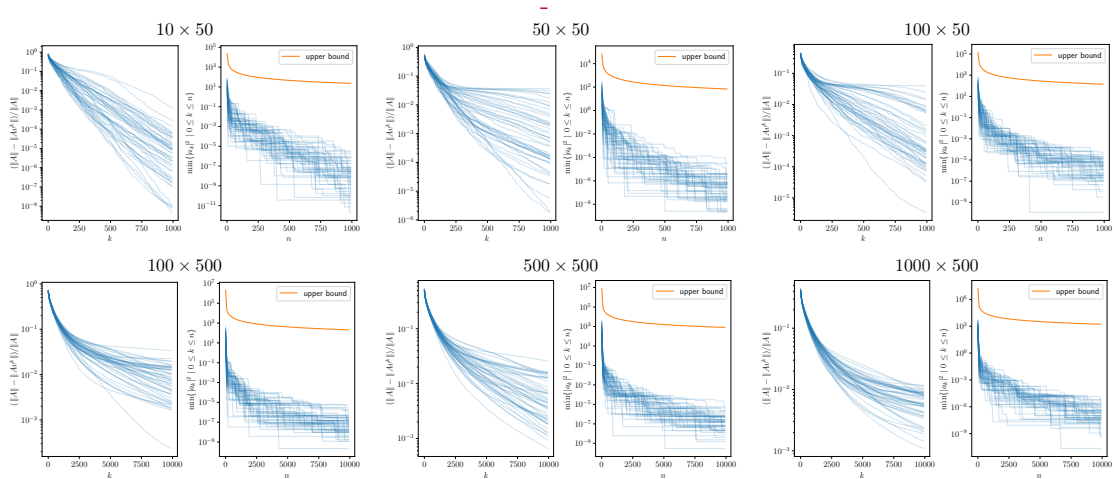


FIGURE 1: Results for 50 runs of our method for Gaussian matrices of different sizes.

RADON TRANSFORMATION

- First: from the Python package `scikit-image`^[a],
 - provide a *filtered* backprojection in `skimage.transform.iradon`
 - when the filter is set to `None`, approximate adjoint is computed.
- Second: from the `odl` Python package^[b],
- **Comparison:**
 - 1) `skimage`: power method on A^*A : $\|Av^k\| \rightarrow 54.84$,
 - 2) `skimage`: power method on A^*A : $\sqrt{\|A^*Av^k\|} \rightarrow 8.22$
 - scaling issue in the adjoint **vs.** our method: 55.86.
 - 3) `odl`: power method 1.9509808 **vs.** our method: 1.9509840.

[a] Stéfan van der Walt et al. “scikit-image: image processing in Python”. In: *PeerJ* 2 (June 2014), e453. DOI: 10.7717/peerj.453.

[b] Frank Natterer and Frank Wübbeling. *Mathematical Methods in Image Reconstruction*. Society for Industrial and Applied Mathematics, 2001. DOI: 10.1137/1.9780898718324.

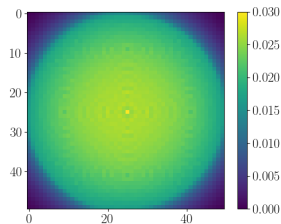
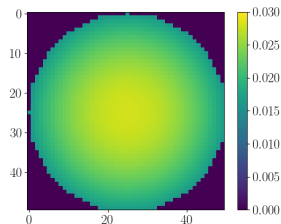


TABLE OF CONTENTS

I. MOTIVATION & CONTRIBUTIONS

II. REMARKS

III. DETAILS

IV. NUMERICAL RESULTS

V. OUTLOOK

- Consider two linear operators A, V between finite dimensional Hilbert spaces.
- Interested in the problem of the adjoint mismatch $\|A - V\|$.
- **Restrictions:** forwards evaluation for A , i.e. $v \mapsto Av$
- and only backward evaluations for V , i.e. $u \mapsto V^*u$ are available.
- **Notice:** $\|A - V\| = \sup_{u,v} \langle u, (A - V)v \rangle = \sup_{u,v} (\langle V^*u, v \rangle - \langle u, Av \rangle)$.
- **Sketch:** construct u^k, v^k again by orthogonal projections w^k, x^k in $(u^k)^\perp$, i.e. $(v^k)^\perp$ with one global, or two optimal stepsizes for each component.
- speed up for two directions; one direction less technical in the proof for the a.s. convergence,
- for two stepsizes, again, the corresponding eigenvalue and eigenvector equation can be studied

$$\min_{0 \leq k \leq n} \|(A - V)^*(A - V)v^k - a_k^2 v^k\|^2 \in \mathcal{O}(n^{-1/2}) \quad \text{a.s.}$$

Thank you for your attention!

REFERENCES

- Kristian Bredies et al. “Degenerate preconditioned proximal point algorithms”. In: *SIAM Journal on Optimization* 32.3 (2022), pp. 2376–2401.
- Antonin Chambolle and Thomas Pock. “A first-order primal-dual algorithm for convex problems with applications to imaging”. In: *Journal of mathematical imaging and vision* 40 (2011), pp. 120–145.
- Weihao Kong and Gregory Valiant. “Spectrum estimation from samples”. In: *Ann. Statist.* 45.5 (2017), pp. 2218–2247. DOI: doi.org/10.1214/16-AOS1525.
- Yi Li, Huy L Nguyen, and David P Woodruff. “On sketching matrix norms and the top singular vector”. In: *Proceedings of the twenty-fifth annual ACM-SIAM symposium on Discrete algorithms*. SIAM. 2014, pp. 1562–1581.
- Frank Natterer and Frank Wübbeling. *Mathematical Methods in Image Reconstruction*. Society for Industrial and Applied Mathematics, 2001. DOI: [10.1137/1.9780898718324](https://doi.org/10.1137/1.9780898718324).
- Ernest K Ryu and Wotao Yin. *Large-scale convex optimization: algorithms & analyses via monotone operators*. Cambridge University Press, 2022.